

Yao Ma, Ph.D.

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Professional Profile

Control and signal-processing engineer with more than ten years of experience developing algorithms for complex dynamical systems and wearable products. Demonstrated ability to translate research into production-quality software, conduct simulation and hardware-in-the-loop experiments, and collaborate across disciplines. Committed to leveraging expertise in robust and optimal control, system identification and signal processing to advance innovative technologies and solve complex problems across various application domains. Record of publication in peer-reviewed journals and conferences.

Core Qualifications

- **Control Systems & Safety Mitigation:** Experience in modeling, stability analysis, robust/optimal control design, model predictive control and reinforcement learning for nonlinear systems; includes design of safety mitigations and evaluation of feature performance in simulation and on hardware.
 - **Signal Processing & Data Analysis:** Acoustics and vibration analysis; filtering and estimation techniques; statistical analysis of experimental data and system logs.
 - **Algorithm Implementation:** Proficient with MATLAB/Simulink, Python and embedded systems; experienced in implementing and testing algorithms on real-time embedded platforms and hardware-in-the-loop systems.
 - **Research Leadership:** Manage multi-year funded research programs; publish in peer-reviewed journals and conferences; mentor engineers and students; document innovations through invention disclosures.
 - **Cross-Functional Collaboration:** Collaborate with DSP, acoustic, hardware, software and testing teams to integrate algorithms into products and ensure compliance with safety and performance requirements.
 - **Project Management:** Develop research plans, manage budgets, write grant proposals and coordinate deliverables to meet critical development timelines.
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Professional Experience

Bose Corporation — Senior Controls System Engineer, Global Research & Development

Framingham, MA | 2023 – Present

- Developed active noise cancellation (ANC) algorithms for smart wearables (headphones, earbuds and hearing aids).
- Performed modeling, control design and real-time implementation of ANC algorithms.
- Analyzed device performance data and conducted tuning and validation of algorithms in prototype and production environments.
- Implemented and tested algorithms on embedded platforms and hardware-in-the-loop systems; ensured stability, safety and low-power operation.
- Collaborated with DSP, acoustic and audio teams to deliver algorithms that meet product requirements and user-experience goals.
- Documented novel ideas and research outcomes in invention disclosures.
- Conducted applied research on **system dynamics, robust and optimal control, signal processing and system identification.**

Texas Tech University — Assistant Professor, Department of Mechanical Engineering

Lubbock, TX | 2019 – 2022

- Taught undergraduate and graduate courses on linear and nonlinear control, optimization and model predictive control.
- Supervised a team of students conducting research in control, autonomous systems and robotics.
- Served as principal investigator for multi-year projects on connected and autonomous vehicles and energy-efficient propulsion systems; designed control and estimation algorithms for hybrid powertrains and vehicle dynamics.
- Secured funding through grant proposals and managed budgets for projects funded by the National Science Foundation and academic partners, with total funding exceeding \$180 k.
- Published 11 journal papers and 16 conference papers (first/corresponding author); organized conference sessions and served as reviewer/editor for leading control journals.
- Collaborated with cross-disciplinary teams to translate research outcomes into practical solutions.

University of Texas at Austin — Graduate Research Assistant, Mobility System Lab

Austin, TX | 2018 – 2019

- Designed control strategies and evaluation metrics for connected and automated vehicles, focusing on energy efficiency and mobility improvement.
- Developed optimization algorithms for vehicle energy and emission performance considering autonomy, connectivity and driver behavior.

The Ohio State University — Graduate Research Associate, Vehicle System & Control Lab

Columbus, OH | 2014 – 2018

- Modeled, controlled and estimated automotive engine, powertrain and after-treatment systems.
- Optimized hybrid vehicle power management and conducted experiments with sensors and data acquisition equipment to inform control algorithm development.

Mohu Consumer Electronics — Electrical Engineer, Product Development

Raleigh, NC | 2013 – 2014

- Designed and prototyped board-level analog circuits for consumer electronics products.
- Collaborated with manufacturing teams on production and quality assurance.

Selected Projects & Research Highlights

- **Smart Wearable Active Noise Cancellation (Bose, 2023 – Present):** Designed robust optimal control algorithms for ANC in wearables; developed intelligent and adaptive algorithms for personalized user environments.
- **Mobility Automation (Texas Tech University, 2019 – 2022):** Principal investigator on projects addressing connected and autonomous vehicles in mixed traffic, driver behavior monitoring and predictive propulsion control; managed a multidisciplinary team and delivered software prototypes and publications.
- **Cyber-Physical System: Synergy (National Science Foundation, 2016 – 2018):** Developed personalized active vehicle safety control using vehicle connectivity technologies; built an interactive driving simulation platform with virtual reality and autonomous steering.
- **Advanced Diesel Engine Aftertreatment System (Tenneco, Inc., 2014 – 2015):** Designed control algorithms for emission control; calibrated and maintained experimental systems (data acquisition, sensors and prototypes).

Education

- **Ph.D., Mechanical Engineering (Control Systems)** — The University of Texas at Austin, 2019
Dissertation: *Synergistic and intelligent control of vehicle powertrain–aftertreatment systems*
 - **M.S., Electrical Engineering (Control & Signal Processing)** — North Carolina State University, 2013
 - **B.S., Control Science and Engineering** — Harbin Institute of Technology, 2012
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Selected Publications

1. Mehmet Fatih Ozkan and Yao Ma, “Distributed Stochastic Model Predictive Control for Human-Leading Heavy-Duty Truck Platoon,” *IEEE Transactions on Intelligent Transportation Systems*, 2022. DOI: 10.1109/TITS.2022.3147719.
2. Mehmet Fatih Ozkan and Yao Ma, “Socially Compatible Control Design of Automated Vehicle in Mixed Traffic,” *IEEE Control Systems Letters*, vol. 6, pp. 1730–1735, 2022. DOI: 10.1109/LCSYS.2021.3133175 — **ASME Automotive and Transportation System Best Paper Award.**
3. Mehmet Fatih Ozkan and Yao Ma, “Eco-Driving of Connected and Automated Vehicle with Preceding Driver Behavior Prediction,” *ASME Journal of Dynamic Systems, Measurement and Control*, Jan 2021, 143(1): 011002. DOI: 10.1115/1.4048108.
4. Yao Ma and Junmin Wang, “Integrated Power Management and Aftertreatment System Control for Hybrid Electric Vehicles with Road Grade Preview,” *IEEE Transactions on Vehicular Technology*, vol. 66, no. 12, pp. 10935–10945, 2017. DOI: 10.1109/TVT.2017.2763587.
5. Mehmet Fatih Ozkan and Yao Ma, “Personalized Adaptive Cruise Control and Impacts on Mixed Traffic,” in *Proceedings of the 2021 American Control Conference*, 2021 — **ASME Automotive and Transportation System Best Paper Finalist.**

Additional publications are listed at drmayao.github.io/publication.

Professional Service & Affiliations

- Member, IEEE Control Systems Society and IEEE Intelligent Transportation Systems Society.

- Reviewer for *IEEE Transactions on Automatic Control*, *IEEE Transactions on Intelligent Transportation Systems* and related journals.
 - Organizer and session chair at the American Control Conference and IEEE CDC.
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Technical Skills

- **Programming & Tools:** MATLAB/Simulink, Python, embedded systems, Git and LaTeX.
 - **Hardware & Platforms:** dSPACE HIL, National Instruments, Analog Devices and real-time microcontrollers.
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Honors & Funding

- **National Science Foundation – CISE (Principal Investigator, 2022):** *Human-Centric Connected and Automated Vehicles for Sustainable Mobility* — \$175 k.
 - **NSF Regional I-Corps Site Program (PI, 2021):** *Real-Time Data Monitoring System with Nanorobotics* — \$3 k.
 - **Texas Tech University – Edward E. Whitacre, Jr. College of Engineering (PI, 2020):** *Alternate Energy Research Initiative* — \$11 k.
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